

Faculty of computer science and information technology

Year 2018/2019

Level:1

Course Name: Calculus 1

Course Code: MAT102

Sheet No.:4

Semester: Fall 2018

Instructor: Dr. RawhyIsmail

Problem 1: Show that

$$i) \cosh x \pm \sinh x = e^{\pm x} \quad ii) \sinh(\ln x) = \frac{x^2 - 1}{2x}$$

$$ii) \cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$iii) \tanh(x + y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y} \quad \text{and hence show that}$$

$$\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$$

$$v) \cosh^2 x + \sinh^2 x = \cosh 2x$$

$$vi) \sinh 2x = 2 \sinh x \cosh x$$

Problem 2: Find the derivative of each of the following functions:

$$(a) y = \sinh 3^x$$

$$(b) y = \cosh \sqrt{3 - e^{-x}} \tanh x^2$$

$$(c) y = \sqrt{\cosh e^{x^2}}$$

$$(d) y = \tanh \left(\frac{e^{-x}}{x^e + x^{-e}} \right)$$

$$(e) y = \operatorname{sech}(\ln(4 - 2x))$$

$$(f) y = e^{\cot h\left(\frac{1}{x}\right)}$$

$$(g) y = \ln(e^{\sqrt{\operatorname{csch}(x)}})$$

$$(h) y = \ln(\operatorname{csch} \sqrt{x})$$

$$(i) y = e^{\ln(\cosh \sqrt{x})}$$

$$(j) y = 4^{\sinh(\cosh 4x)}$$

$$k) y = \frac{1}{\sqrt[3]{\tanh x e}}$$

Problem 3 : Solve the following equations (round the result to 3 decimal places):

i) $\cosh(2\sqrt{x} + 1) = 3$ ✗ ii) $\coth\left(\frac{x-1}{x+1}\right) = \frac{5}{3}$

iii) $\sinh^{-1}(e^x) = 1.2$ v) $\operatorname{sech}^{-1}(\ln x) = 2$

Problem 4: Find all the hyperbolic values of x given that

a) $\operatorname{sech} x = \frac{3}{4}$

b) $\coth x = \frac{4}{3}$

Problem 5: Without using a calculator, evaluate

a) $\sinh^{-1} 1/2$

b) $\tanh^{-1} 1/2$
